

Questionnaire and Study Measures

Local Trait Inference on Smartphones

DRAFT – NOT FOR SUBMISSION OR DATA COLLECTION

Administrative and data-management fields marked “TO CONFIRM” require supervisor approval.

This appendix lists all participant-facing questionnaire items planned for the Android study. Scale values are stored numerically together with their verbal labels. The optional camera module is a data-collection instrument, not a questionnaire.

A. Eligibility and Demographic Questions

A1. Age confirmation I confirm that I am at least 18 years old.

Options: Yes / No. A “No” response ends eligibility.

A2. Age group Which age group are you in?

Options: 18–24; 25–34; 35–44; 45–54; 55 or older; prefer not to say.

A3. Gender How do you describe your gender?

Options: woman; man; non-binary; self-description; prefer not to say.

A4. Highest completed education level What is the highest level of education you have completed?

Options: school qualification; vocational training; bachelor’s degree; master’s degree or higher; other; prefer not to say.

Implementation alignment note: The current prototype lacks “prefer not to say” for age, and lacks “self-description”/“other” options in two items. These options should be aligned before data collection.

B. Internet Users’ Information Privacy Concerns – IUIPC-8

Instruction: The following statements concern online privacy. Please indicate how strongly you agree or disagree with each statement.

Response scale: 1 strongly disagree; 2 disagree; 3 somewhat disagree; 4 neither agree nor disagree; 5 somewhat agree; 6 agree; 7 strongly agree.

Control

B1. Consumer online privacy is really a matter of consumers’ right to exercise control and autonomy over decisions about how their information is collected, used, and shared.

B2. Consumer control of personal information lies at the heart of consumer privacy.

Awareness

B3. Companies seeking information online should disclose the way the data are collected, processed, and used.

B4. A good consumer online privacy policy should have a clear and conspicuous disclosure.

Collection

B5. It usually bothers me when online companies ask me for personal information.

B6. When online companies ask me for personal information, I sometimes think twice before providing it.

B7. It bothers me to give personal information to so many online companies.

B8. I am concerned that online companies are collecting too much personal information about me.

Scoring note for researchers: Preserve the three dimensions Control, Awareness, and Collection. Confirm the planned scoring and psychometric analysis with the psychology supervisor before preregistration or final analysis.

Source: Gross, T. (2023). Toward Valid and Reliable Privacy Concern Scales: The Example of IUIPC-8. In *Human Factors in Privacy Research*. The eight-item form omits ctrl3 and awa3 from IUIPC-10 and uses a seven-point agreement scale.

C. Affinity for Technology Interaction – ATI

Instruction: In the following questionnaire, we will ask you about your interaction with technical systems. The term “technical systems” refers to apps and other software applications, as well as entire digital devices such as a mobile phone, computer, TV, or car navigation system.

Response scale: 1 completely disagree; 2 largely disagree; 3 slightly disagree; 4 slightly agree; 5 largely agree; 6 completely agree.

- C1. I like to occupy myself in greater detail with technical systems.
- C2. I like testing the functions of new technical systems.
- C3. I predominantly deal with technical systems because I have to.
- C4. When I have a new technical system in front of me, I try it out intensively.
- C5. I enjoy spending time becoming acquainted with a new technical system.
- C6. It is enough for me that a technical system works; I do not care how or why.
- C7. I try to understand how a technical system exactly works.
- C8. It is enough for me to know the basic functions of a technical system.
- C9. I try to make full use of the capabilities of a technical system.

Scoring note for researchers: Items C3, C6, and C8 are reverse-coded. Compute the mean across all nine items after reversal. Report the mean, standard deviation, and internal consistency as appropriate.

Source: Franke, T., Attig, C., & Wessel, D. (2019). A Personal Resource for Technology Interaction: Development and Validation of the Affinity for Technology Interaction (ATI) Scale. *International Journal of Human-Computer Interaction*, 35(6), 456–467. <https://doi.org/10.1080/10447318.2018.1456150>

D. Baseline Local-AI Expectations

Instruction: Please indicate how strongly you agree or disagree before seeing any file analysis.

Response scale: 1 strongly disagree; 2 disagree; 3 neither agree nor disagree; 4 agree; 5 strongly agree.

- D1. I expect a phone to infer sensitive information from selected local files.
- D2. The statement “data stays on the phone” makes this feel privacy-friendly.
- D3. I usually enjoy trying new digital technologies.
- D4. I am concerned about how apps can use information stored on my phone.

Design note: D3 overlaps conceptually with ATI and D4 overlaps with IUIPC-8. Confirm whether they are needed as brief face-valid items or should be removed to reduce burden and avoid redundant measurement.

E. Post-Inference Questions

These three questions are shown after each successfully analyzed image or PDF first-page preview. The displayed probabilistic inference remains visible while the participant responds.

Response scale: 1 not at all; 2 slightly; 3 moderately; 4 strongly; 5 very strongly.

- E1. How unexpected was this inference?
- E2. How sensitive does this inference feel?
- E3. How accurate does this inference seem?

Each response is linked to a randomized file identifier and its model output. Original filenames and local paths are excluded from the final research export.

F. Combined Profile and Final Questions

Before these questions, participants see one compact probabilistic personal profile synthesized from the file-level inferences. The following framing text is displayed:

Imagine a normal app performed this kind of local analysis regularly in the background and only sent inferred summaries or profile categories away.

Response scale: 1 strongly disagree; 2 disagree; 3 neither agree nor disagree; 4 agree; 5 strongly agree.

F1. Before this study, I expected my phone to be able to run this kind of AI analysis locally.

F2. I would be concerned if an app repeated this analysis regularly on local photos, screenshots, or documents.

F3. The statement “the original files never leave the device” would still reassure me if inferred summaries or profile categories were sent to a server.

F4. I would be comfortable if an app analyzed local files in the background and sent inferred summaries to a server without clearly notifying me each time.

F5. I would be comfortable if this inferred profile were used commercially, for example for advertisements, product offers, or sharing with another company.

F6. After seeing this profile, I would still allow broad photo or file access to an app.

G. Optional Front-Camera Reaction Measure

This is not a self-report questionnaire. After separate opt-in, the app may sample a low-resolution front-camera frame approximately every 12 seconds while the relevant study interface is active. Local classification yields:

- timestamp;
- study-screen event;
- model identifier/version;
- predicted label from angry, disgust, fear, happy, neutral, sad, or surprise; and
- model confidence.

No audio is captured. The final implementation must delete each temporary frame immediately and retain no image bytes or file path. This measure is exploratory and must not be interpreted as a direct measure of the participant’s true emotional state or as a diagnosis.

H. End-of-Study Information

Participants are reminded that:

- all displayed traits are probabilistic model outputs and may be wrong;
- the original selected files were processed locally and were not part of the research export;
- local processing does not by itself prevent privacy impact if derived summaries or profiles are transmitted; and
- they may contact the study team with questions or request deletion using their study ID until the stated cutoff.